Code No: 114AG **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, May - 2019
FORMAL LANGUAGES AND AUTOMATA THEORY
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART – A

(25 Marks)

1.a) Define DFA and NFA. [2]
b) Design NFA to accept set of all strings over {0,1} does not contain 3 consecutive zeros. [3]
c) Define Left Linear Grammar. [2]
d) Define Pumping Lemma for Regular sets. [3]
e) Define UNIT Production. [2]
f) What is reachable variable? Give an example. [3]
g) Give instantaneous description ID of Turing Machine. [2]
h) What is a Turing Machine? [3]
i) Give any two examples of Decidable Problems. [2]
j) Define NP hard problems. [3]

PART – B

(50 Marks)

2.a) Minimizing the following DFA as shown in figure 1.

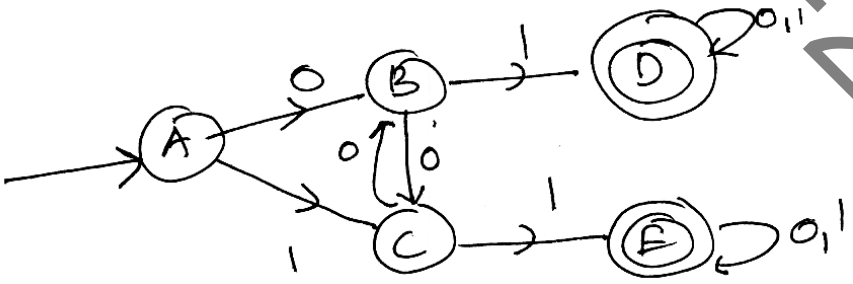


Figure: 1

b) Define Mealy and Moore Machines. [6+4]

OR

3.a) Convert the following NFA with € - moves to DFA as shown in Figure 2.

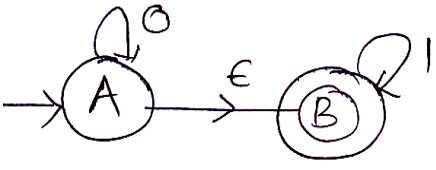


Figure: 2

b) Design DFA for the language all strings over {0,1} which are ending with 00**.** **[**5+5]

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4.a) Show that the language L={ $a^n b^n$ / n≥1 } is not a Regular language.

b) Find the Regular Grammar equivalent to the following Finite Automata shown in figure 3. [5+5]

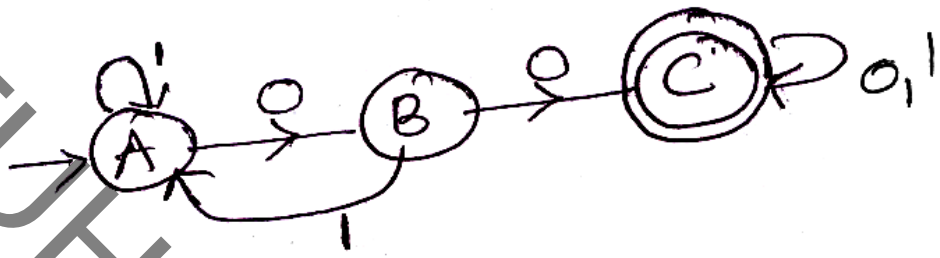


Figure 3

OR

5.a) Construct Right Linear Grammar and Left Linear Grammar for the Regular Expression $0^*(11)^*+1001^*$

b) Give the RE for
(i) All strings over {a,b} which contains even number of a's
(ii) All strings over {a,b} which contains at least two a's
(iii) All strings over {a,b} which contains at most two a's
(iv) All strings over {a,b} which contains exactly two a's. [6+4]

6.a) Minimize the following Context Free Grammar G=({S,A,B,C}, {a,b,c,d},P,S), where P is given as follows
S→AC | SB
A→ bASC | a
B→ aSB | bbB
C→ Bc | ad

b) Design PDA for the language L={$a^{2n}b^nc^4$/n≥1}. [5+5]

OR

7.a) Convert the following Context Free Grammar (CFG) to Chomskey Normal Form (CNF)
S→AaB | aaB
A→€
B→bbA | €

b) Design PDA for the language L={$a^n b^{n+m} c^m$ | n,m≥1}. [5+5]

8.a) Explain different types of Turing Machine.

b) Design TM for performing proper subtraction of two numbers. [5+5]

OR

9. Write about:
a) Church's hypothesis.
b) Counter machine. [5+5]

10.a) Discuss about turing reducibility.

b) What is post correspondence problem? Verify whether the following PCP has solution or not? A={ba,ab,a,baa,b}, B={bab,baa,ba,a,aba} [5+5]

OR

11. Define the NP-complete problems and give examples. [10]

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